7 Experiences Summit 2025
Around the World in 48 Hours (Online)
September 29-30, 2025

Segmentation of Cruise Experience Preferences: A Hierarchical Cluster Analysis of Budget-Constrained Choices

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Abstract

This study investigates cruise experience preferences through a hierarchical cluster analysis of budget-constrained choices. The research identifies three distinct passenger segments by surveying potential cruise passengers: Experience-Focused Luxury Cruisers, Entertainment and Relaxation Seekers, and Active Explorer Cruisers. Each segment demonstrates unique spending patterns and prioritization of cruise experience dimensions, such as cabin preferences, service expectations, food and beverage options, relaxation activities, and entertainment. The findings reveal that Experience-Focused Luxury Cruisers prioritize premium drinks and luxury experiences; Entertainment and relaxation Seekers focus on onboard entertainment and relaxation, while Active Explorer Cruisers emphasize premium service, high-quality food, and sightseeing opportunities. This segmentation provides valuable insights for cruise operators to optimize service offerings and marketing strategies, effectively enhancing customer satisfaction and targeting specific passenger segments. The study highlights the importance of understanding passenger preferences under budget constraints to inform strategic service design in the cruise industry.

Keywords

customer experience, cruise, passenger, preference, cluster, segment

Introduction

This study focuses on the cruise experience, a form of tourism experience in a moving vessel, usually on circular routes, returning to the departure port after visiting several weekly destinations. Racid (2018, 135) defines the nature of the cruise experience as follows: **"**The cruise experience is a personal, highly emotional, memorable event that is created on the cruise ship and visited ports of call involving multi-sensory interactions between service providers and guests, guests and service areas, and between the guests themselves." While experiences differ according to a person's earlier experiences, motivation, culture, and many other attributes, cruise ships must fulfil different passenger needs. The need for fulfilment has become even more crucial as cruise ships have evolved from a means of transportation between destinations towards ships becoming destinations per se. Significant changes, like the service and expanded passenger segments, highlight the importance of considering different passenger segments' expectations for the cruise experience.

Cruising is a significant part of the tourism industry, with a passenger volume of 31.7 million in 2023 (CLIA, 2024), and the cruise experience has many definitions. According to Haievska (2023), modern cruises are a unique phenomenon in the tourism sector because they combine elements of different types of tourism (recreational, educational, sightseeing, and beach), and in cruises, everything is designed for the tourist's pleasure, and their relaxing (accommodation, food, and leisure). Ahola, Salovuori, and Lehtonen, 2015) explored cruising through social constructivism to find how human and environmental factors affect the cruising experience and concluded that the cruise experience is ordered from everyday distinctions, predictability, and social experiencing. Racid (2018) divides the cruise experience into four building blocks: *Service theatre* (employees, guests, and service setting), *Service quality* (safety, courtesy, show, and efficiency), *The experience realms* (education, escapism, aesthetics, and entertainment), and *Experienced emotions.* Naturally, different passenger segments value building blocks of the cruise experience in various ways, and more research is needed to identify common patterns in the perceived value of the cruise experience. This study addresses this need for understanding by clustering passenger preferences.

Chua et al. (2015) discovered that the perceived ticket price negatively impacts the perceived value. In contrast, interaction and overall quality relate positively to novelty and perceived value. Qu and Ping's (1999) study concluded that the cruise experience arising from escapism, aesthetics, entertainment, and socialization significantly influences the passengers' motivation and overall satisfaction with their cruise vacation. The highest level of comfort and the possibility of combining several types of tourism have become a symbol of modern cruise tourism (Haievska, 2023). A substantial part of this development is caused by improved socio-cultural support (ibid.). This study highlights the relationship between ticket prices and expected experiences. In other words, how passengers are willing to invest or compromise the expected experience if money is a determinant factor.

Background

Cruise Passenger Preferences

The analysis of cruise ship passenger preferences and profiles is a multi-faceted area of research that integrates, for example, behavioral science, marketing dynamics, and tourism trends. Understanding these dimensions is critical for cruise companies striving to enhance customer satisfaction and optimize service offerings. Papathanassis (2024) offers a broader perspective on cruise tourism research and its evolution. Highlighting the historical shift from a ship-centric to a more passenger-oriented approach, this synthesis calls attention to the necessity for ongoing research that focuses on operational aspects and prioritizes understanding the diverse profiles of cruise passengers. By envisioning the future of cruise tourism, the study advocates for sustainable practices that consider passenger expectations and experiences as central elements in the industry's growth (Papathanassis, 2024).

We discovered several different categorizations for the critical attributes of the cruise experience, from which the most relevant top-level categories to our study are collected in Table 1. Beyond purchasing behavior, perceived value has been identified as a critical determinant in cruise passengers' overall experience and behavioral intentions. Baker and Fulford (2016) found that perceived value explained approximately 45% of the variation in passengers' willingness to recommend a Caribbean cruise to others. Their research identified five factors that significantly contributed to cruise passengers' perceived value: (1) overall satisfaction with the cruise experience, (2) quality of service received, (3) consistency of cruise arrangements with what was promised, (4) food quality aboard the ship, and (5) relaxing nature of the island destinations. Together, these five factors explained 81% of the variability in perceived value. Notably, including destination factors among these key determinants suggests that cruise passengers value destinations that offer relaxation opportunities, highlighting the importance of considering both onboard and off-board experiences in passenger preference analysis.

In a recent study, Demydyuk and Carlbäck (2024) identified several critical factors driving overall cruise satisfaction across different market segments. Their research revealed that perceived value for money emerged as the strongest predictor of overall satisfaction across all market segments (mainstream, premium, luxury, and ultra-luxury), with the highest coefficient observed in the mainstream segment. Core hospitality elements, including dining, service, cabin quality, and onboard activities, significantly contributed to perceived value and ship quality perceptions. Notably, service quality had particularly strong effects in the luxury segment, while in mainstream cruises, all experience attributes contributed more equally to satisfaction. The researchers also identified important moderating variables that influenced satisfaction relationships. Price and cabin category corresponded with elevated expectations, with higher cabin categories in the mainstream market showing a positive moderation effect on the cabin-ship quality relationship. Group size also affected satisfaction levels, with larger travel groups in the mainstream market showing higher overall satisfaction but more critical evaluations of value and ship quality. Previous cruising experience consistently led to more critical evaluations across most satisfaction metrics, particularly in mainstream and premium markets (Demydyuk & Carlbäck, 2024).

Dining experiences represent a crucial dimension of cruise passenger preferences, appearing in most of the identified categories (See Table 1.). Demydyuk et al. (2024) applied the Experience Accounting (EA) framework to analyze cruise dining preferences, categorizing dining experiences into four key "experience accounts": basic food and beverage (F&B), culinary finesse, service, and restaurant atmosphere. Their research identified significant post-pandemic shifts in passenger preferences, including movement from basic food requirements toward fine-dining options, greater emphasis on culinary finesse and variety, increased attention to food quality in complementary dining venues, and growing interest in vegetarian, vegan, and protein options. The study also identified distinct F&B venues that elicit different passenger preferences, including main dining rooms, buffet restaurants, room service, specialty restaurants, and bars.

Sentiment analysis methodologies provide valuable insights into more detailed cruise passenger preferences. Sun et al. (2023) analyzed over 97,000 online reviews covering 172 cruise ships across 16 brands to extract sentiment patterns from user-generated content. Their research identified 10 key cruise attributes that influence passenger satisfaction and sentiment: dining (26.7% of cruise characteristic words), entertainment (15.2%), cabin (13.5%), service (10.9%), embarkation (6.7%), value for money (4.8%), shore excursions (4.5%), fitness and recreation (2.1%), public rooms (1.6%), and enrichment activities (0.9%). The study also identified distinct patterns in how passengers express sentiment about cruise experiences, with positive sentiment drivers including enrichment activities, service, entertainment, fitness and recreation, and shore excursions. In contrast, negative sentiment drivers included embarkation processes, service issues, cabin concerns, dining experiences, and entertainment shortcomings (ibid).

Service quality perceptions play a critical role in understanding cruise passenger preferences. Arasli et al. (2020) conducted a critical content analysis of 2,000 online reviews to identify service quality dimensions that shape cruise travelers' perceptions. Their research identified ten major themes in service quality perception: ship, staff, food, entertainment, room/cabins, public spaces/areas, embarkation process, excursions, disembarkation process, and port. Furthermore, Arasli et al. (2020) found distinct preference patterns between high-satisfaction and low-satisfaction passenger groups. Passengers who rated their cruise experience highly (4-5 out of 5) predominantly valued ship quality and aesthetics, public spaces and facilities, room comfort and amenities, staff helpfulness and friendliness, food quality and variety, and entertainment offerings. In contrast, passengers who rated their cruise experience poorly (1-2 out of 5) were more concerned with process-oriented aspects, including embarkation/disembarkation inefficiencies, excursion disappointments, port issues, reservation problems, cancellations, and perceived value issues related to expense relative to experience.

Several key factors that influence cruise selection and pricing have been identified, which can serve as valuable variables for passenger profiling (Espinet-Rius et al., 2018): itinerary characteristics, cabin type, ship characteristics, onboard amenities, and food and beverage options.

Scholars have lately been interested in Asian markets due to cruise operators' interest in tailoring their services according to Asian preferences for cruise experiences. Tao and Kim (2019)

researched Asian cruise passenger experiences, employing big data analytics to examine online cruise reviews and identify factors influencing passenger satisfaction in the Asian market. Their factor analysis identified six key dimensions of cruise experience that explained 40% of the variance in passenger satisfaction: Onshore Attributes, Dining, Value, Core Products, Staff, and Entertainment.

Chang et al. (2021) employed a choice experiment (CE) method to identify five key attributes that influence cruise tourists' preferences in the Taiwanese market (See Table 1). Their research quantified passengers' marginal willingness to pay (MWTP) for these attributes, revealing the highest willingness to pay for sea-view cabins with balconies, followed by Michelin restaurant experiences, with varying WTP for refund mechanisms.

Kong et al. (2022) identified several key service dimensions influencing Chinese outbound cruise tourists' satisfaction. Their study developed three categories of service attributes, including physical environment quality (cabin interiors, entertainment facilities, accommodation quality), food and dining experience (quality and selection of foods/dishes), staff interaction quality (communication skills, attitude, service responsiveness), and process quality (boarding services, ticket handling, complaint management). Their Importance-Performance Analysis (IPA) revealed that attributes like comfortable cabin interiors, public spaces, and staff attitude were highly valued and well-performed, while safety perceptions, food quality/variety, and staff communication skills showed significant gaps (Kong et al., 2022).

Beyond traditional factors, technological infrastructure now plays a significant role in shaping cruise passenger preferences. González-Santiago et al. (2024) identify several technology-related factors that impact passenger experience and preferences. Smart wearable devices such as Princess Cruises' Ocean Medallion and MSC's "MSC for Me" wristbands not only serve practical functions (cabin keys, contactless payments) but also influence how passengers interact with and experience cruise services (Buhalis et al., 2022; Kwee-Meier et al., 2016). Additionally, cruise applications have become standard across major cruise lines, allowing passengers to "make purchases, order drinks, and book excursions or dinners" (Saltzman & Garay, 2020), which impacts their purchasing preferences and decision-making processes during the cruise experience.

Table 1. Identified customer experience attributes within the literature review.

Attribute	Key cruise attributes (Sun et al. (2023)	Service quality dimensions shaping cruise travelers' perceptions (Arasli et al., 2020)	Key factors of cruise selection and pricing (Espinet-Rius et al., 2018)	Critical factors for cruise satisfaction (Demydyuk & Carlbäck, 2024a)	Perceived value factors (Baker & Fulford, 2016)	Factors influencing Asian passenger satisfaction (Tao & Kim, 2019)	Key attributes influencing cruise tourists' preferences in the Taiwanese market (Chang et al., 2021)	Key service dimensions for Chinese cruise satisfaction (Kong et al., 2022)	Technology related factors in cruise services (González-Santiago et al., 2024)
Food / beverage / dining	X	X	X	X	X	X	X	X	
Onshore attributes	X	X	X		X	X			
Cabin	X	X	X	X			X	X	
Ship services	X	X	X	X		X			
Entertainment	X	X		X		X	X		
Public areas	X	X	X				X		
Embarkation	X	X							
Other				X			X		X
Categorization method	Sentiment analysis	Convenience sampling	Multiple Correspondence Analysis (MCA)	Text-mining and Experience Accounting	Passenger cruise experience questionnaire	Web scraping of online reviews	Choice Experiment (CE)	Importance-Performance Analysis (IPA)	Literature review

Categorization name and reference

Furthermore, the purchasing behavior of cruise passengers has been significantly influenced by cruise attributes, as explored by Kwon and Ahn (2021). Their findings indicate that attributes such as the variety of selections and competitive pricing affect the initial decision to purchase and stimulate impulse buying tendencies among consumers. This suggests an intricate relationship between cruise attributes and passengers' emotional and psychological triggers influencing purchasing behaviors during the cruise experience (Kwon & Ahn, 2021). The study provides insights into how specific marketing strategies can be designed based on customer profiles, particularly highlighting the impact of promotional activities onboard. They establish a framework combining personality dimensions with emotional responses, indicating that passenger profiles characterized by high impulsivity are more likely to engage in unplanned purchases (Ahn & Kwon, 2020).

Beyond the specific cruise experience elements, nationality has emerged as a significant factor influencing visitor satisfaction and behavioral intentions in tourism contexts. Vieira et al. (2021) empirically demonstrated that nationality is a crucial variable affecting satisfaction levels and loyalty (measured through willingness to recommend), even when controlling for demographic and trip-related factors.

The behavioral intentions of cruise passengers have been a focal point in recent studies. For instance, López-Marfil et al. (2021) examined senior tourists' specific preferences and behaviors regarding cruise ports of call. Their research highlights how factors such as emotional and cultural connections seniors have with destinations influence their decisions and intentions to engage with specific cruise services, emphasizing the importance of demographic characteristics in passenger

profiling (López-Marfil et al., 2021). This reflects a shift towards personalized service offerings, where understanding the customer's identity and expectations is increasingly vital.

Beyond traditional factors and technology adoption, comprehensive data collection frameworks can enhance our understanding of cruise passenger preferences. Espinet et al. (2024) propose a structured approach that distinguishes between ports of embarkation/disembarkation and ports of call when analyzing passenger behaviors. The research identifies several critical data categories for understanding passenger preferences, including socio-demographic characteristics, behavioral and motivational attributes, origin and destination patterns, and activity and expenditure tracking during port visits (Espinet et al., 2024). Moreover, Rungroueng and Monpanthong (2023) have identified five key dimensions that significantly influence cruise passengers' perceptions of port quality, providing valuable insights into passenger preferences: Cruise Terminal Facilities, Port Service Encounter Performance, Port Location, Ground Transportation of the Port, and Port Physical Environment Quality. Their more recent research by Rungroueng (2024) provides valuable insights into the crucial attributes that influence cruise passenger satisfaction, particularly related to port experiences. The study identified five key port attributes that significantly affect passenger satisfaction and behavioral intentions: Port Service Provider, Port Accessibility, Port Environment, Passenger Port Building, and Passenger Transportation in Port. These findings highlight that the physical environment of ports has the strongest influence on passenger perceptions, followed closely by transportation options and accessibility features (Rungroueng, 2024).

Integrating multiple data sources can provide a more comprehensive view of passenger preferences. Espinet et al. (2024) suggest utilizing shipping company records, survey data (particularly at embarkation points), customer experience ratings from platforms like cruisecritic.com, web scraping of online reviews (Tao & Kim, 2019), and social media data analysis (Park et al., 2016). These diverse data sources can reveal patterns in comparative passenger profiles, behavioral segmentation, impact analysis, and preferences related to seasonality and extended stays (Espinet et al., 2024).

The ASCT framework proposed by González-Santiago et al. (2024) highlights technology readiness as a critical moderating variable affecting how passengers interact with onboard technologies. Wang et al. (2017) and Yoganathan et al. (2021) suggest that passenger technology readiness significantly influences satisfaction levels and overall experience, offering a key dimension in understanding passenger preferences. This technological aspect complements the traditional understanding of cruise preferences, suggesting that technology acceptance now forms an integral part of the modern cruise experience.

While most previous studies have identified the different preference factors, fewer have focused on the financial constraints involved in customer decision-making. A study by Neuts, Chen, and Nijkamp (2016) offers a comprehensive approach to understanding customer value in cruise markets through a combination of willingness-to-pay (WTP) and probability of repeat cruising. Their research identified several significant variables that affect customer value and are particularly relevant for market segmentation: socio-demographic factors (with age and income showing positive relationships with customer value), previous cruise experience (strongly

predictive of future value), preference for cruise characteristics (longer cruises correlated with higher value), and cruise motivations (specifically the "escaping" motivation showed a significant positive relationship with customer value). This integrated approach addresses limitations in traditional customer valuation methods by combining both current spending patterns and future intentions, providing a more holistic understanding of cruise passenger preferences (Neuts et al., 2016).

Penco and Di Vaio (2014) determined that passenger expenditures are influenced by weather conditions, hours spent ashore, and demographic/socioeconomic characteristics. Building on this, Ferrante et al. (2018) discovered that passengers with higher incomes, higher education levels, and ages between 36-55 years tend to seek more intense destination experiences. These insights into passenger behavior and expenditure patterns offer important dimensions for effective passenger profiling and cruise service design (Machado da Luz et al., 2022). Understanding differences between preference patterns requires suitable methods to profile the cruise customers.

Cruise Passenger Profiling Methods

There are several ways to profile cruise ship passengers and to understand the distinction between the different types of passengers traveling with cruise ships. Rungroueng and Monpanthong (2023) employed Generalized Structured Component Analysis (GSCA), which revealed both formative and reflective relationships between perceived port quality and its constituent dimensions. The strong correlations between perceived port quality and passenger satisfaction indicate that clusters based on port quality preferences may also predict satisfaction levels and behavioral intentions (Rungroueng and Monpanthong, 2023).

Another notable approach to cruise passenger profiling comes from Neuts et al. (2016), who employed latent cluster analysis (LCA) for market segmentation in Japanese and Taiwanese cruise markets. Their analysis identified six distinct market segments: Habitual cruisers (mature travelers with prior cruise experience, preferring longer cruises but showing low motivation levels), Inexperienced younger cruisers (below 40 years old, lower income, minimal cruise experience, with varying motivation patterns), Short cruise enthusiasts (young travelers, lower income, minimal experience, preferring shorter cruises but high on all motivation categories), Experienced senior cruisers (over 60 years old, higher income, extensive cruise experience, preferring longer cruises, with varied motivation patterns). Experienced senior cruisers demonstrated the highest customer value, highlighting the importance of age, experience, and motivation factors in identifying high-value market segments.

Complementing these segmentation approaches, Baker and Fulford (2016) examined cruise passenger profiles through the lens of the theory of planned behavior (TPB) (Fishbein & Ajzen, 1975), which proposes that behavior is influenced by behavioral intentions, affected by attitudes and subjective norms. Their research captured important demographic and behavioral characteristics that provide valuable segmentation variables for clustering analysis, including geographic origin, gender distribution, age distribution, and previous cruise experience. Their methodological approach, which utilized real-time assessment from actual passengers during an ongoing cruise rather than retrospective evaluation, enhances the validity.

Another approach to cruise passenger profiling is presented by Espinet et al. (2024), who emphasize the value of big data and business intelligence in developing nuanced passenger profiles. Their framework distinguishes between "native" passengers (those living near embarkation/disembarkation ports) and "tourists" (those traveling from elsewhere), noting different behavioral patterns and economic impacts between these groups. For effective cruise passenger profiling, Espinet et al. (2024) highlight the importance of capturing comprehensive socio-demographic data, including age, gender, academic training, professional activity, and family income range. Beyond demographics, understanding the motivational and behavioral characteristics of the cruise passenger during the itinerary is essential, including motivations for taking the cruise and previous cruise experience. Vieira et al. (2021) underscore the importance of nationality as a distinctive segmentation criterion in tourism contexts. Their research identified several variables with significant explanatory power that could enhance cruise passenger clustering models. Beyond the fundamental nationality variable, they highlighted the value of measuring expectation disconfirmation (the gap between expectations and perceived experience) as a strong predictor of satisfaction. Their analysis incorporated various trip-related characteristics that demonstrated meaningful explanatory power, including participation in specific leisure activities, a season of travel, whether it was a first visit, destination choice ranking, accommodation preferences, and travel party composition. Interestingly, while their model controlled for traditional demographic variables such as education, gender, marital status, occupational status, and age, many of these showed limited explanatory value compared to nationality and trip-specific factors, suggesting that behavioral and experiential variables may offer more meaningful segmentation criteria than standard demographics alone.

Complementing these profiling approaches, Demydyuk and Carlbäck (2024) demonstrated that the cruise market can be effectively segmented into distinct categories (mainstream, premium, luxury, and ultra-luxury), each with unique customer expectations and satisfaction drivers. While perceived value for money was universally important across all segments, the weight of other satisfaction factors varied significantly between these segments.

Service perception patterns offer another promising approach to cruise passenger profiling. In their content analysis of cruise reviews, Arasli et al. (2020) revealed apparent differences between high-satisfaction and low-satisfaction passenger groups based on the service aspects they prioritized. The high-satisfaction group demonstrated a likelihood of discussing public spaces, entertainment, staff helpfulness, and food quality. Meanwhile, the low-satisfaction group frequently mentioned expenses, reservation issues, long lines, elevator problems, and cancellations. These variations in service element prioritization could serve as valuable variables for developing more refined passenger profiles through hierarchical clustering. The distinction between passengers prioritizing tangible service elements versus those focusing more on intangible service processes adds another dimension for segmentation that complements the market segment divisions (mainstream, premium, luxury, ultra-luxury) identified by Demydyuk and Carlbäck (2024).

Market positioning differentiation represents another valuable dimension for cruise passenger profiling. Sun et al. (2023) identified significant differences in passenger sentiment expressions

across market segments (standard/contemporary, premium, and luxury cruises). Their findings show that differences between brand positionings, illustrated by tourists' positive and negative sentiments, suggesting that market segmentation is a crucial factor in passenger profiling. This sentiment-based approach to distinguishing passenger profiles across market segments aligns with the findings of Demydyuk and Carlbäck (2024) regarding distinct satisfaction drivers across market segments, while adding the dimension of sentiment expression patterns as a potential clustering variable.

Rationale and Aim

Despite extensive research on cruise passenger preferences, there remains a significant gap in understanding how passengers prioritize different aspects of the cruise experience when faced with budget constraints. While previous studies have examined isolated preference dimensions or post-experience satisfaction (e.g., Sun et al., 2023; Demydyuk & Carlbäck, 2024), limited research has investigated the trade-offs passengers make across multiple experience dimensions simultaneously when resources are finite.

This gap is particularly relevant given the cruise industry's continuous expansion and increasing service diversification. As cruise ships evolve from transportation vessels to destinations themselves, understanding how different passenger segments allocate limited resources across competing experiential offerings becomes critical for strategic service design and targeted marketing efforts.

The literature has identified numerous factors influencing cruise satisfaction, including dining quality, entertainment options, service interactions, and physical environment (Kong et al., 2022; Tao & Kim, 2019). However, most studies have examined these factors in isolation or through post-cruise satisfaction ratings rather than exploring pre-purchase decision trade-offs. Market segmentation efforts have identified distinct passenger types (Neuts et al., 2016; Demydyuk & Carlbäck, 2024), but few have examined how these segments differ in their preferential budget allocations across multiple experience dimensions.

This study addresses these gaps by examining how prospective cruise passengers allocate a finite budget across eight key experiential dimensions: cabin preferences, service expectations, beverage options, food preferences, relaxation activities, experiential offerings, sightseeing options, and onboard entertainment. By imposing realistic financial constraints (a 3000 euro budget), the research creates conditions that more accurately reflect real-world consumer decision-making processes where trade-offs are inevitable. The experiential dimensions relate with the categories presented in Table 1. Out of the categories identified by the previous publications, we have selected the categories relevant to the during cruise experience. Thus, we are not focusing on the pre- and post-experiential dimensions such as embarkment. Furthermore, we have chosen the categories in that a customer can select while booking the cruise. Therefore, we are not focusing on e.g., the public space, as the customer cannot choose the quality of the public space during the booking process.

The study aims to answer the following research questions:

RQ1 What distinct passenger segments emerge when cruise consumers must allocate a finite budget across multiple experiential dimensions?

RQ2 How do these segments differ in their prioritization of and willingness to pay for specific cruise experience elements?

RQ3 What patterns of preference trade-offs characterize each segment when financial resources are constrained?

By employing hierarchical cluster analysis with budget-constrained preference data, this research provides a more nuanced understanding of cruise market segmentation than approaches based solely on demographics or unconstrained preference ratings. The findings offer practical insights for cruise operators seeking to optimize service offerings, pricing strategies, and marketing communications for specific passenger segments based on their demonstrated willingness to invest in particular experiential dimensions over others.

Methods

Participants and Data Collection

The study analyzed preference data from Finnish potential cruise passengers (N=28) who voluntarily participated in the survey. No demographic information was collected from participants, representing a limitation that should be addressed in future studies (Dolnicar, 2020).

Participants were all Finnish citizens whose relationship with cruising is exceptional since in Finland, it is extremely common to travel with cruise ferries on the Baltic Seas. The cruise ferries sailing in the Baltic Sea are imitating the cruise ship's service scape and experience within a shorter sailing duration and route, simplified amenities, and lack of sundeck. To our knowledge, the Finns' preferences for cruises have not been the research subject of segmentation studies.

Data was collected using a structured survey that presented respondents with three options across eight key dimensions of cruise experiences: cabin preferences, service expectations, beverage options (drinks), food preferences, relaxation activities, experiential offerings, sightseeing options, and onboard entertainment. Each option within a category varied in quality and price, reflecting real-world trade-offs cruise travelers face (Petrick, 2004). Each dimension offered three tiered options with corresponding price points (e.g., interior cabin [€1500], window cabin [€1900], balcony cabin [€2500]), allowing for measurement of both categorical preferences and willingness to pay.

To create the conditions for examining RQ3 (patterns of preference trade-offs under financial constraints), participants were allocated an imaginary budget of 3000 euros to distribute across their cruise preferences. This budget constraint was deliberately set so that selecting the highest-quality option in all categories would exceed the total available funds (total cost of all premium

options: €4350), forcing participants to prioritize certain aspects of their cruise experience over others (Jacoby & Chestnut, 1978). This constraint directly enabled the analysis of spending passenger segments and preference trade-offs presented in the Results section.

Data Preprocessing

For each preference dimension, responses were processed to create two types of variables necessary for the analyses presented in the Results: (1) categorical variables representing the verbal preference selection (e.g., "balcony cabin") and (2) corresponding numerical values representing the monetary allocation and strength of preference (e.g., €2500).

To address the potential impact of different price scales on the clustering results (cabin prices ranged from €1500-2500 while drink package prices ranged from €0-300), all numerical variables were z-score normalized prior to clustering analysis (Hair et al., 2019). This standardization, essential for the cluster identification presented in the Results, ensured that variables with more extensive price ranges would not disproportionately influence the clustering algorithm. The standardized values were used for the cluster analysis and profile plots, while the original monetary values were retained to calculate the total expenditure distributions.

Hierarchical Cluster Analysis

To answer RQ1 regarding identifying distinct passenger segments, the analysis employed hierarchical clustering using Ward's minimum variance method with Euclidean distance as the similarity metric. This method was selected because it tends to create compact, evenly-sized clusters by minimizing the total within-cluster variance (Murtagh & Legendre, 2014), making it appropriate for market segmentation studies where the optimal number of clusters is unknown a priori (Everitt et al., 2011).

The clustering algorithm was applied to the eight normalized preference variables across all 28 respondents. This produced the hierarchical structure that yielded the three-cluster solution described in the Results. To validate this solution, we also conducted K-means clustering with k=3, which produced identical cluster memberships, providing methodological triangulation for the segment structure.

The optimal number of clusters was determined through visual inspection of the dendrogram structure and statistical validation via the Gap statistic method, which compares the total within intra-cluster variation for different values of k with their expected values under the null reference distribution of the data. This produced the Gap statistic plot, showing an inflection point at 3 clusters that statistically supported our three-segment solution.

Statistical Validation

To validate the distinctiveness of the identified clusters and address RQ2, a one-way analysis of variance (ANOVA) was conducted for each of the eight preference variables across the three clusters. For variables demonstrating significant overall differences ($p < 0.05$), Tukey's Honest Significant Difference (HSD) post-hoc tests were performed to identify specific between-cluster

differences, following recommendations by Field (2013). These post-hoc tests produced pairwise comparison results, identifying which cluster pairs differed significantly on each preference dimension. Additionally, to quantify the structure of the solution, we calculated the within-cluster sum of squares (WCSS) values and the between-clusters sum of squares to confirm the statistical validity of the three-cluster structure.

To analyze the specific categorical preference patterns within each cluster, frequency analyses were conducted for each preference category across the three identified segments. Chi-square tests were used to examine whether categorical preference distributions (e.g., choices between balcony, window, or interior cabins) differed significantly between clusters. For spending pattern analysis (RQ3), descriptive statistics including means, medians, and standard deviations were calculated for each preference category by cluster. These calculations produced the spending distribution patterns, which showed the total expenditure distributions for each segment.

Multiple visualization techniques were employed to represent the cluster characteristics and differences, following recommendations by Kassambara (2017). The visualization suite included cluster dendrograms to illustrate the hierarchical structure of the segmentation, heatmaps displaying preference intensities across variables for each cluster, and profile plots showing standardized mean scores for each cluster across all preference dimensions. Additionally, box plots and confidence interval plots were created to illustrate the distribution of scores and between-cluster differences, while frequency distribution charts visualized categorical preferences by cluster. All statistical analyses were performed using the Jamovi statistical software (Version 2.6) with the snowCluster module (Version 7.4.6) and supporting R packages, including factoextra for visualization of multivariate analyses (Kassambara & Mundt, 2020).

Results

Cluster Identification and Validation

The hierarchical cluster analysis using Ward's minimum variance method yielded a three-cluster solution representing distinct cruise passenger segments. This solution was independently validated through K-means clustering, which produced identical cluster assignments (4, 17, and 7 members, respectively), providing robust triangulation of the segmentation structure.

The optimal number of clusters was determined by examining the Gap statistic (Figure 1), which demonstrated an inflection point at 3 clusters, statistically supporting the three-segment solution. The within-cluster sum of squares values (9.21, 74.71, and 50.85 for clusters 1, 2, and 3 respectively) and a between-clusters sum of squares of 81.23 further confirmed the appropriateness of the three-cluster structure.

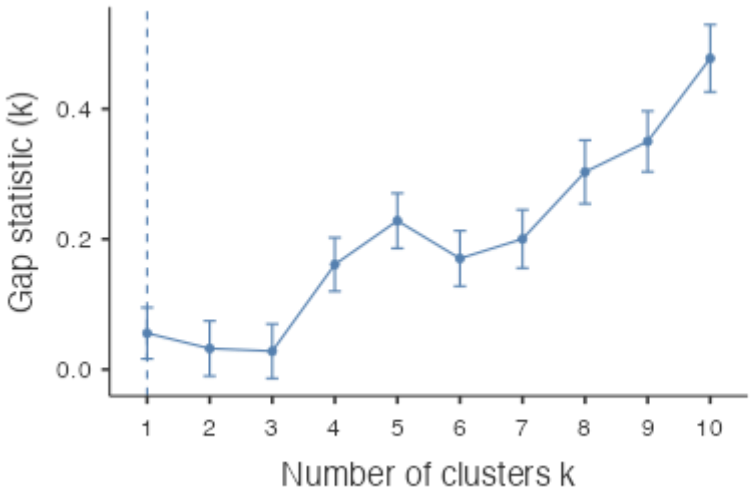


Figure 1. Gap statistics showing the optimal solution for clustering.

One-way ANOVA analyses (Table 2) revealed statistically significant differences between clusters across seven of the eight measured preference dimensions ($p < .05$): service ($F(2,25) = 11.211$, $p < .001$), drinks ($F(2,25) = 7.774$, $p = .002$), food ($F(2,25) = 3.474$, $p = .047$), relaxing ($F(2,25) = 5.819$, $p = .008$), experiences ($F(2,25) = 25.560$, $p < .001$), sightseeing ($F(2,25) = 7.506$, $p = .003$), and entertainment at ship ($F(2,25) = 12.530$, $p < .001$). Only cabin preferences did not demonstrate significant between-cluster variation ($F(2,25) = 0.940$, $p = .404$), suggesting similar cabin priorities across all passenger segments.

Table 2. Results of the One-Way ANOVA analysis between clusters.

	F	df1	df2	p
Cabin	0.940	2	25	0.404
Service	11.211	2	25	<.001
Drinks	7.774	2	25	0.002
Food	3.474	2	25	0.047
Relaxing	5.819	2	25	0.008
Experiences	25.560	2	25	<.001
Sightseeing	7.506	2	25	0.003
Entertainment	12.530	2	25	<.001

Post-hoc Tukey tests (Table 3) identified specific between-group differences, with particularly strong distinctions in preferences for experiences, service quality, and entertainment options. The standardized preference profile plot clearly illustrates these divergent patterns, with each cluster demonstrating a distinctive preference signature across the measured variables.

Table 3. The Tukey Post-Hoc Test Results between significant preference categories.

Tukey Post-Hoc Test – Service

		1	2	3
1	Mean difference	—	8.82	−136 *
	t-value	—	0.229	−3.13
	df	—	25.0	25.0
	p-value	—	0.971	0.012
2	Mean difference		—	−145 ***
	t-value		—	−4.65
	df		—	25.0
	p-value		—	<.001
3	Mean difference			—
	t-value			—
	df			—
	p-value			—

Note. * p < .05, ** p < .01, *** p < .001

Tukey Post-Hoc Test – Relaxing

		1	2	3
1	Mean difference	—	8.82	35.7 *
	t-value	—	0.804	2.89
	df	—	25.0	25.0
	p-value	—	0.704	0.021
2	Mean difference		—	26.9 *
	t-value		—	3.03
	df		—	25.0
	p-value		—	0.015
3	Mean difference			—
	t-value			—
	df			—
	p-value			—

Note. * p < .05, ** p < .01, *** p < .001

Tukey Post-Hoc Test – Sightseeing

		1	2	3
1	Mean difference	—	−26.5	−114.3 **
	t-value	—	−0.846	−3.24
	df	—	25.0	25.0
	p-value	—	0.678	0.009
2	Mean difference		—	−87.8 **
	t-value		—	−3.48
	df		—	25.0
	p-value		—	0.005
3	Mean difference			—
	t-value			—
	df			—
	p-value			—

Note. * p < .05, ** p < .01, *** p < .001

Tukey Post-Hoc Test – Drinks

		1	2	3
1	Mean difference	—	118 *	200.0 **
	t-value	—	2.61	3.93
	df	—	25.0	25.0
	p-value	—	0.039	0.002
2	Mean difference		—	82.4
	t-value		—	2.26
	df		—	25.0
	p-value		—	0.081
3	Mean difference			—
	t-value			—
	df			—
	p-value			—

Note. * p < .05, ** p < .01, *** p < .001

Tukey Post-Hoc Test – Experiences

		1	2	3
1	Mean difference	—	350 ***	92.9
	t-value	—	5.88	1.38
	df	—	25.0	25.0
	p-value	—	<.001	0.365
2	Mean difference		—	−257.1 ***
	t-value		—	−5.34
	df		—	25.0
	p-value		—	<.001
3	Mean difference			—
	t-value			—
	df			—
	p-value			—

Note. * p < .05, ** p < .01, *** p < .001

Tukey Post-Hoc Test – EntertainmentAtShip

		1	2	3
1	Mean difference	—	−88.2 ***	−85.71 ***
	t-value	—	−4.90	−4.223
	df	—	25.0	25.0
	p-value	—	<.001	<.001
2	Mean difference		—	2.52
	t-value		—	0.173
	df		—	25.0
	p-value		—	0.984
3	Mean difference			—
	t-value			—
	df			—
	p-value			—

Note. * p < .05, ** p < .01, *** p < .001

Cluster Characteristics and Spending Patterns

Based on the observed preference patterns, the three clusters were labeled as follows: Experience-Focused Luxury Cruisers (n = 4, 14.3%), Entertainment & Relaxation Seekers (n = 17, 60.7%), Active Explorer Cruisers (n = 7, 25.0%). The differences between cluster profiles are illustrated in Figure 2.

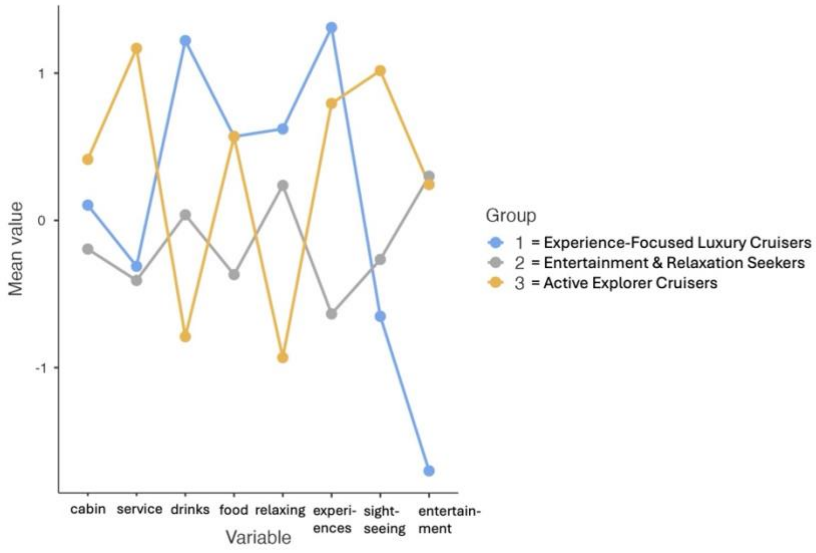


Figure 2. Customer profiles. X axis presents the preference categories, and Y axis represents the normalized mean values per cluster.

Experience-Focused Luxury Cruisers segment demonstrated the strongest preference for premium drinks (M = 300.0, SD = 0.0) and experiential offerings (M = 350.0, SD = 173.2), with a moderate interest in cabin quality (M = 1900.0, SD = 115.5) and high-quality food (M = 200.0, SD = 115.5). These passengers consistently selected spa facilities for relaxation (100%) and showed a distinct lack of interest in onboard entertainment (M = 0.0, SD = 0.0) and shore excursions (M = 0.0, SD = 0.0). The standardized profile plot (z-scores) showed particularly high values for drinks (z = 1.221), experiences (z = 1.310), and relaxing (z = 0.621), with notably low values for entertainment (z = -1.701). This segment demonstrated relatively high total expenditure (M = 2850, median = 2850), suggesting a willingness to pay premium prices for specific luxury experiences.

Entertainment & Relaxation Seekers segment allocated moderate budgets across multiple preference categories, representing the majority of the sample, with notable spending on entertainment at ship (M = 88.2, SD = 33.2) and moderate expenditure on drinks (M = 182.4, SD = 101.5). These passengers showed limited interest in experiential offerings (M = 0.0, SD = 0.0) and shore excursions (M = 26.5, SD = 25.7), with moderate investment in relaxation facilities (M = 91.2, SD = 19.6). The standardized profile revealed moderate positive values for entertainment (z = 0.300) and relaxing (z = 0.237), with negative values for experiences (z = -0.635) and food (z = -0.368). This segment showed the lowest total expenditure (M = 2371, median = 2400), suggesting value-conscious cruisers who prioritize onboard entertainment and relaxation.

Active Explorer Cruisers segment demonstrated strong preferences for premium service (M = 185.7, SD = 106.9), high-quality food (M = 200.0, SD = 129.1), and sightseeing opportunities (M = 114.3, SD = 106.9). These passengers invested substantially in experiential offerings (M = 257.1, SD = 181.3) but allocated a minimal budget to drinks (M = 100.0, SD = 0.0) and relaxation facilities (M = 64.3, SD = 24.4). The standardized profile showed strong positive values for service (z = 1.168), sightseeing (z = 1.017), and experiences (z = 0.794), with notably negative values for drinks (z = -0.790) and relaxing (z = -0.932). This segment showed the highest total expenditure (M = 2893, median = 2950), suggesting travelers willing to invest in active, exploration-oriented cruise experiences. In Figure 3, the spending distribution patterns between segments are presented.

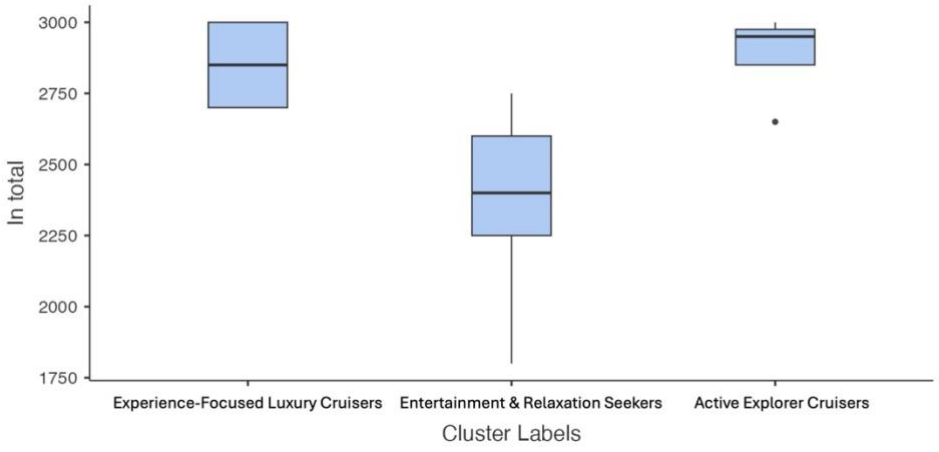


Figure 3. Cluster spending patterns: x axis presents the clusters, and y axis presents the total money spend distribution per cluster.

Preference Patterns Across Categories

Categorical preference distributions provided additional insights into each segment's priorities. While statistical tests indicated no significant difference in cabin budget allocation, the categorical preference analysis revealed varying selection patterns. Active Explorers showed the strongest preference for balcony cabins (71%) compared to Entertainment & Relaxation Seekers (47%) and Experience-Focused Luxury Cruisers (50%). Experience-Focused Luxury Cruisers (50%) and Entertainment & Relaxation Seekers (35%) selected window cabins more frequently. Active Explorers distinctly prioritized the captain's reception services (43%) compared to other segments (0%), while Experience-Focused Luxury Cruisers and Entertainment & Relaxation Seekers predominantly selected cleaning services (50% and 59% respectively). All Experience-Focused Luxury Cruisers (100%) selected the all-inclusive drinks package. In comparison, Active Explorers unanimously chose soft drinks packages (100%), and Entertainment & Relaxation Seekers showed a mixed preference pattern (41% all-inclusive, 59% soft drinks).

Active Explorers demonstrated the strongest preference for exclusive dining options (57%), while Experience-Focused Luxury Cruisers equally divided between fine dining and exclusive options (50% each). Entertainment & Relaxation Seekers showed the most frequent selection of buffet options (29%). All Experience-Focused Luxury Cruisers (100%) selected spa facilities, while Entertainment & Relaxation Seekers also strongly preferred spa options (82%). Active Explorers showed the highest preference for pool facilities (71%) compared to other segments. Active Explorers strongly preferred scuba diving excursions (57%), while Experience-Focused Luxury Cruisers divided equally between scuba and safari options (50% each). For onboard entertainment, both Entertainment & Relaxation Seekers and Active Explorers overwhelmingly preferred dinner and show options (88% and 86%, respectively), while Experience-Focused Luxury Cruisers unanimously selected movies (100%). These distinctive preference patterns across categories validate the three-segment solution and provide valuable insights for developing targeted cruise service offerings and marketing strategies.

Discussion

Summary of Key Findings

The present study explored how prospective cruise passengers allocate a finite budget across multiple experiential dimensions, revealing three distinct passenger segments with unique preference patterns and spending priorities. Through hierarchical cluster analysis, the research identified Experience-Focused Luxury Cruisers (14.3%), Entertainment & Relaxation Seekers (60.7%), and Active Explorer Cruisers (25.0%) as distinct market segments.

The Experience-Focused Luxury Cruisers demonstrated pronounced preferences for premium drinks packages (consistently selecting all-inclusive options) and luxury experiential offerings while showing minimal interest in onboard entertainment and shore excursions. This segment unanimously selected spa facilities for relaxation and displayed moderate interest in cabin quality, predominantly choosing window or balcony accommodations. Their average total expenditure

(€2850) approached the maximum budget constraint, indicating a willingness to invest substantially in specific luxury dimensions of the cruise experience.

Entertainment & Relaxation Seekers, representing the majority of the sample, allocated moderate budgets across multiple preference categories with a notable emphasis on onboard entertainment (predominantly selecting dinner shows) and relaxation facilities (primarily spa options). This segment showed limited interest in experiential offerings and shore excursions, with mixed preferences for drink packages. Their total expenditure was the lowest among the three segments (€2371), suggesting a value-conscious approach focused on entertainment and relaxation dimensions.

Active Explorer Cruisers exhibited the strongest preferences for premium service (particularly captain's reception), high-quality food (predominantly exclusive dining), and sightseeing opportunities (with emphasis on scuba diving excursions). This segment demonstrated a substantial investment in experiential offerings while allocating a minimal budget to drinks (unanimously selecting soft drinks packages) and relaxation facilities (primarily pool options). Their total expenditure was the highest (€2893), reflecting their willingness to invest in destination exploration and premium service experiences.

Comparison with Previous Research

The identified segments demonstrate alignment with and divergence from previous cruise market segmentation research. The Experience-Focused Luxury Cruisers segment aligns conceptually with the "habitual cruisers" identified by Neuts et al. (2016), particularly in their preference for premium experiences and higher expenditure patterns. However, while Neuts et al. found this segment to represent relatively mature travelers (over 60 years), demographic data limitations in the current study prevent similar age-related comparisons.

The Active Explorer Cruisers segment shares characteristics with what Sun et al. (2023) identified as passengers whose positive sentiment was driven by enrichment activities, shore excursions, and service quality. This segment's willingness to allocate a substantial budget to experiential offerings and sightseeing aligns with Chang et al.'s (2021) finding that Taiwanese cruise tourists demonstrated a significant willingness to pay for premium experiences. The segment's strong preference for high-quality food and service also corresponds with Demydyuk and Carlbäck's (2024) observation that service quality particularly strongly affected satisfaction in the luxury segment.

The Entertainment & Relaxation Seekers segment parallels several previous findings in the literature. Their moderate allocation across multiple categories, with emphasis on entertainment and relaxation, resonates with what Racid (2018) identified as the "entertainment" and "escapism" realms of cruise experience. This segment's price sensitivity aligns with Chua et al.'s (2015) finding that perceived ticket price significantly impacts perceived value, and their preference pattern suggests what Baker and Fulford (2016) identified as valuing the "relaxing nature" of the cruise experience.

Interestingly, all three segments showed some degree of preference for higher-quality cabin accommodations, with no significant between-segment differences in cabin budget allocation. This finding diverges somewhat from Chang et al.'s (2021) research, which found cabin comfort (particularly sea-view cabins with balconies) to be the attribute with the highest willingness to pay among Taiwanese cruise tourists. This divergence may reflect cultural differences or be attributable to the study's specific budget constraint methodology.

The pattern of spending differences across the three segments aligns with Demydyuk and Carlbäck's (2024) finding that different market segments (mainstream, premium, luxury) demonstrate varying satisfaction drivers and experiential priorities. The Experience-Focused Luxury Cruisers and Active Explorer Cruisers segments, with their higher total expenditures, show parallels to Demydyuk and Carlbäck's premium and luxury segments, while the Entertainment & Relaxation Seekers segment, with lower expenditure and more moderate preferences, resembles their mainstream segment characterization.

The distinctive preference patterns for relaxation facilities, with Experience-Focused Luxury Cruisers unanimously selecting spa options while Active Explorers predominantly chose pool facilities, adds nuance to Kong et al.'s (2022) findings regarding physical environment quality preferences among Chinese cruise tourists. Similarly, the significant variation in beverage package preferences across segments (from 100% all-inclusive in Experience-Focused Luxury Cruisers to 100% soft drinks only in Active Explorers) provides valuable insight beyond Demydyuk et al.'s (2024) general observation of growing post-pandemic interest in fine-dining and beverage options.

The budget allocation methodology employed in this study offers a novel perspective on preference trade-offs not directly captured in previous research, which has primarily focused on post-experience satisfaction ratings (Arasli et al., 2020; Sun et al., 2023) or willingness to pay for isolated attributes (Chang et al., 2021). By forcing participants to prioritize spending within a finite budget, the research reveals what passengers value and what they are willing to sacrifice when resources are constrained, providing complementary insights to existing literature.

While this study segments passengers based on their budget allocation preferences, it is crucial to acknowledge that these profiles may not represent fixed personality traits. As noted by Falk (2009), visitor identities are often fluid and situational rather than static. A passenger who falls into the Entertainment & Relaxation Seeker segment when traveling with young children may shift to the Experience-Focused Luxury Cruiser identity when traveling as a couple on an anniversary trip. Therefore, the segments identified in this study should be viewed as "behavioral modes" adopted for a specific trip purpose, rather than permanent classifications of the individuals. This fluidity suggests that cruise operators should not only profile customers based on past behavior but also assess the specific context of their current booking to offer relevant upgrades and experiences.

Managerial Implications

The identification of these three distinct segments offers actionable insights for cruise operators to tailor their marketing communications and service design. To attract Experience-Focused Luxury Cruisers, marketing efforts should move beyond mass-market channels. Operators should

utilize niche, high-end travel publications and direct marketing that emphasizes exclusivity, privacy, and the all-inclusive nature of the "ship-within-a-ship" concept. Visuals for this group should focus on the tranquility of the spa and the quality of premium beverages, rather than high-energy activities.

Conversely, the Entertainment & Relaxation Seekers—the largest segment—are likely best reached through broader digital campaigns and social media platforms where value bundles can be highlighted. Promotional strategies for this group should focus on the "fun factor," showcasing dinner shows and pool areas, while clearly communicating the value proposition of standard cabins combined with inclusive entertainment.

For Active Explorer Cruisers, marketing narratives must pivot to the destination and the culinary journey. Operators should partner with adventure travel portals and emphasize the itinerary's unique ports of call and culinary excellence. Marketing materials should frame the ship as a gateway to exploration rather than just a floating resort, highlighting the quality of excursions and dining over beverage packages.

Furthermore, as suggested by the rationing preferences observed in this study, these findings have implications for strategic demarketing. In destinations suffering from overtourism or environmental strain, cruise operators could adjust their pricing structures to favor the "high-yield, low-footprint" behavior of the Experience-Focused Luxury Cruisers, while effectively pricing out segments that contribute to congestion without significant economic contribution. By manipulating the cost of specific attribute bundles (e.g., raising the price of basic shore excursions while bundling premium experiences), operators can curate a passenger mix that aligns with sustainable tourism goals.

Limitations

Several limitations must be acknowledged when interpreting the findings of this study. First and foremost, the sample size (N=28) is relatively small for cluster analysis, which limits statistical power and the generalizability of the results. While the identified clusters demonstrated statistical validity through multiple validation methods (Gap statistic, ANOVA, and K-means triangulation), a larger sample would provide more robust segment identification and potentially reveal additional sub-segments not captured in the current analysis.

Second, the study's sample composition represents a significant limitation. All participants were Finnish, creating a homogeneous sample lacking diversity in nationality, age range, income level, and prior cruise experience. Given Vieira et al.'s (2021) finding that nationality significantly influences tourism satisfaction and behavioral intentions, the results may not be generalizable to international cruise markets. Additionally, as Dolnicar (2020) has emphasized, the absence of demographic information collection further constrains the ability to characterize the identified segments beyond their preference patterns.

Third, while providing valuable insights into preference trade-offs, the artificial budget constraint methodology represents a hypothetical decision-making scenario that may not perfectly mirror

real-world cruise purchase behaviors. The 3000 euro budget cap was selected to force prioritization but may not reflect the actual budget ranges of diverse cruise market segments. Additionally, while designed to represent realistic trade-offs, the categorical choice options within each preference dimension necessarily simplify the complex, multi-attribute decision-making process involved in actual cruise purchases.

Fourth, the study focused exclusively on pre-purchase preferences rather than incorporating post-experience satisfaction metrics. As Petrick (2004) has noted, perceived value encompasses both pre-purchase expectations and post-experience evaluations. The absence of longitudinal data connecting preference allocations to subsequent satisfaction limits the study's ability to validate whether the identified preference patterns predict cruise satisfaction outcomes.

Finally, the cluster imbalance (4, 17, and 7 members) raises questions about whether this distribution accurately represents the actual market structure or is a potential methodological artifact of the small sample size. The Entertainment & Relaxation Seekers segment's dominant representation (60.7%) may not reflect its true proportion in the broader cruise market.

Future Research

Future research should address the limitations of this study while expanding its methodological approach in several directions. First, a larger, more diverse sample that includes varied nationalities, age groups, income levels, and prior cruise experience would enhance the generalizability of the segmentation results. In particular, collecting comprehensive demographic data alongside preference allocations would enable more nuanced segment profiling, as Espinet et al. (2024) recommended.

Second, extending the research to include post-purchase satisfaction measures would provide a more comprehensive understanding of how preference allocations translate to cruise experience outcomes. A longitudinal design that tracks passengers from budget allocation through actual cruise experiences to post-cruise satisfaction would be particularly valuable for validating the predictive utility of the segmentation approach.

Third, future research could explore the relationship between budget allocation patterns and broader travel motivations as conceptualized in the theory of planned behavior (Fishbein & Ajzen, 1975). Incorporating measures of attitudinal preferences, subjective norms, and perceived behavioral control alongside budget allocations could provide a more theoretically grounded understanding of the psychological drivers behind the identified segments.

Finally, integrating the budget allocation methodology with conjoint analysis techniques could provide more precise estimates of the relative utility values passengers place on different cruise attributes. This more sophisticated approach would enable cruise operators to optimize pricing strategies and service bundling for each identified segment, potentially enhancing customer satisfaction and revenue optimization.

Conclusions

This study contributes to the cruise tourism literature by examining preference patterns and spending allocations of potential cruise passengers when faced with budget constraints. The findings provide valuable insights into how cruise passengers make trade-offs when financial resources are constrained. These segmentation results have significant theoretical implications. The study extends previous research by moving beyond post-experience satisfaction metrics to examine pre-purchase budget allocation patterns, offering a more nuanced understanding of preference formation under resource constraints. For cruise operators, the practical implications are substantial. Understanding these distinct passenger segments enables more targeted service design and marketing strategies. Cruise companies can develop tiered service packages tailored to each segment's preference patterns.

In conclusion, this study demonstrates that cruise passengers make distinctive preference trade-offs when allocating finite budgets across multiple experiential dimensions. The three identified segments represent meaningful market segments with unique spending priorities and service expectations. By understanding these preference patterns, cruise operators can develop more targeted service offerings and marketing strategies, potentially enhancing both customer satisfaction and revenue optimization in an increasingly competitive cruise industry. While budget constraints may limit individual passengers' choices, they reveal valuable insights into the true priorities that drive cruise experience preferences.

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